

CODE SPORT



In this month's column, we feature a set of computer science interview questions.

Last month, we had discussed a set of questions on data science, algorithms and machine learning. We continue this discussion in this month's column, focusing on machine learning and natural language processing.

1. We are all familiar with spoken language recognition systems, which can perform speech-to-text (STT) recognition. Systems such as Apple's Siri, Google Now and Microsoft's Cortana have to deal with a number of complexities as different speakers have widely varying accents and use different styles of pronunciation. Let us consider a person who is reciting the poem: 'Twinkle, Twinkle, Little Star, How I wonder what you are'. Let us assume that the STT system has, so far, correctly identified all the words spoken, except for the last word 'are'. You are given access to a large corpus of English words on which the STS system can be trained. How would you design an algorithm that can recognise the poem completely, assuming that it has been able to correctly recognise all previous words in that poem spoken so far?
2. Can you explain what are unigrams, bigrams and trigrams? How can you use these n-grams to build a language model which can predict the next word in a sentence spoken in English? Though English has a huge vocabulary, let us just assume that we are dealing with a limited vocabulary of 1,000,000 words. For this vocabulary size, how many parameters need to be estimated for the language model?
3. Let us again consider the STS system mentioned in Question (1), above. We are given a sentence spoken in English and have been asked to convert it to text. For example, consider the sentence, "I took a flight to New York." Let us assume that the name 'New York' did not occur at all in the training corpus you have used to train the STS system. However, the training corpus did contain the sentence, "I took a flight to Paris." Can you explain how your language model will recognise the name 'New York'?
4. You are given the task of writing a program that can recognise automobile number plates. The number plate is of the form XXNN-XXNNNN, where X stands for an upper case English letter of the alphabet (between A-Z) and N stands for a single digit between 0-9.
 - a. You have access to a regular expression package available from your favourite programming language. So all you need is to provide the appropriate regular expression to the regular expression library. What is the regular expression that you would come up with?
 - b. You are told that you can't use any pre-existing regular expression library. You are asked to write C code to recognise whether an input is a valid number plate expression or not.
5. We are familiar with spelling and word completion suggestions that appear as we type a search query in various search engines and the suggestions they provide when we misspell a phrase. Can you design a system which takes the user input as a character stream and provide on-the-fly spelling correction suggestions and auto-completion of phrases for the English language? Now, if you are asked to implement this system for a different language like Hindi, is it possible to adapt your existing system to the new language? If yes, explain how you would do that? If not, explain why it is not possible.
6. We are all familiar with Wikipedia articles. Let us assume that you are given access to all the Wikipedia articles that were created in the last two years. You are asked to find out what subjects are popular in Wikipedia, i.e., articles on those subjects get edited often. What NLP technique would you